

Learning from Its Own Footprint: Strategic Disclosure, Capital Feedback, and the Design of AI Valuation Rules in IPO Allocation

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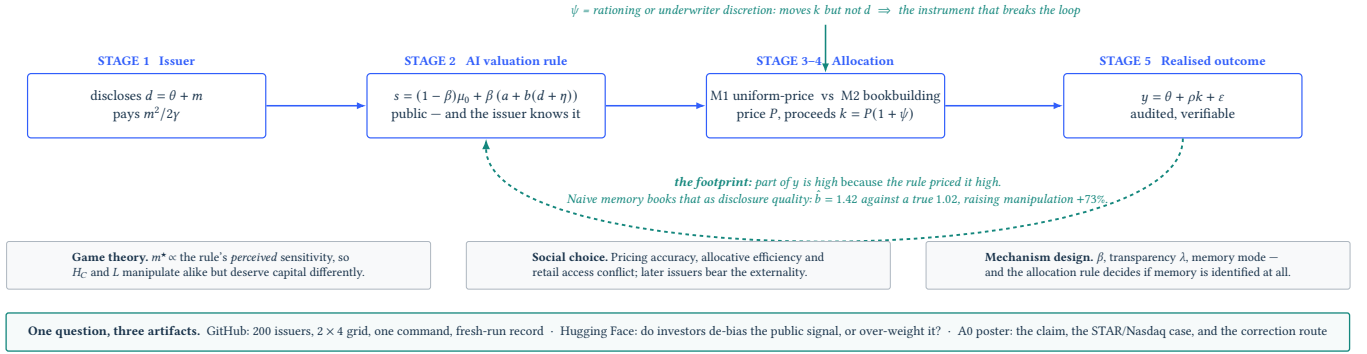


Figure 1: The loop this paper studies. The issuer manages the disclosure the rule reads; the rule's output affects capital allocation; capital changes realised outcomes; and those outcomes can enter the rule's later history. The current simulation reports a higher learned sensitivity and more manipulation under naive memory. Breaking the feedback requires variation in proceeds that is plausibly orthogonal to disclosure. Each lens below changes the analysis rather than merely labelling it.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Team:** FP3. **Symposium:** [Your assigned session].

Course instructor: Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

1 Research question and verified gap

AI-assisted equity valuation creates a strategic environment: an issuer chooses what to disclose, an evaluator converts that disclosure into a public valuation signal, and investors use the signal when allocating scarce capital. The issuer knows the evaluator's rule, so disclosure is potentially strategic. The deeper problem begins when the evaluator learns from later firm outcomes. If its own valuation affects how much capital a firm receives, and capital affects later performance, then part of the evaluator's training data is generated by the evaluator itself.

Shared question. When firms know how an AI evaluator scores their disclosures, can a history-aware evaluator and an auction-based capital-allocation mechanism reduce strategic disclosure while preserving informative pricing?

We answer this through three connected questions. **Q1—strategic behavior:** how does a firm respond when it knows the evaluator and the evaluator's output affects capital? **Q2—mechanism and computation:** can evaluator learning separate genuine firm quality from performance caused by its own allocation? **Q3—behavior:** do investors remove predictable disclosure inflation from the AI signal, as the equilibrium benchmark assumes?

Verified gap. Costly-disclosure models study strategic reporting under a fixed evaluation rule [8, 9]. IPO research studies uniform-price auctions and bookbuilding as information and allocation mechanisms [10–12]. Performativity work studies how deployed predictors change the environment they predict [13]. These literatures do not jointly model a predictor that both affects capital allocation and later learns from outcomes partly caused by that allocation. Our bounded gap is therefore the feedback loop, not the generic claim that “AI matters in finance.”

2 Strategic model and solution concept

There are sequential issuers $t = 1, \dots, N$, investors, and a mechanism designer. Issuer t privately knows fundamental quality θ_t , private value of capital v_t , and social marginal product of capital ρ_t . It chooses disclosure

$$d_t = \theta_t + m_t, \quad c(m_t) = m_t^2/(2\gamma). \quad (1)$$

The evaluator publishes

$$s_t = (1 - \beta)\mu_0 + \beta[a + b(d_t + \eta_t)], \quad (2)$$

where b is the learned disclosure sensitivity. Investors observe s_t and private signals $x_{it} = \theta_t + \xi_{it}$, then bid. The allocation rule determines price P_t and proceeds k_t . Finally,

$$y_t = \theta_t + \rho_t k_t + \epsilon_t \quad (3)$$

becomes part of the history used for later calibration.

The information structure is incomplete because θ_t, v_t, ρ_t are private types, and imperfect because manipulation m_t is not directly observed. The game is sequential, so we use **Perfect Bayesian**

Equilibrium [2]. Under the quadratic cost and single-crossing condition, the firm’s best response is increasing in the evaluator’s perceived sensitivity:

$$m_t^* = \gamma v_t n \lambda_P [\lambda \beta b + (1 - \lambda) \kappa_0]. \quad (4)$$

Thus revealing the evaluation rule changes behavior. The benchmark prediction is not that firms literally fool a fully rational evaluator: investors can anticipate average inflation. The economic cost can instead appear as resources spent to optimize for the machine.

Challenge test. If a history-aware evaluator learns a larger b from outcomes it partly caused, then the best response above increases even when fundamental quality has not changed. This creates a falsifiable prediction: naive history should increase manipulation relative to a no-memory baseline.

3 Social choice and design objective

The social planner does not simply maximize AI prediction accuracy. The relevant stakeholders are issuers, institutional investors, retail investors, exchanges/regulators, and later issuers who inherit the evaluator’s calibration. We therefore track three competing objectives:

- (1) **Informative pricing:** minimize error between market price and latent fundamental value.
- (2) **Allocative efficiency:** direct scarce capital toward firms with high social marginal product ρ .
- (3) **Manipulation and access:** reduce disclosure-management waste while preserving meaningful retail participation.

These objectives can conflict. A mechanism that produces strong institutional information may reduce retail access; a transparent evaluator may improve accountability while also making the target easier to manipulate. The project therefore reports a trade-off rather than a single “best” rule.

The feedback loop also creates an intertemporal externality: one issuer’s strategic disclosure can change the evaluator’s future calibration for later issuers. A socially acceptable evaluator should therefore distinguish genuine information from performance induced by its own previous allocations.

4 Mechanism design and auction comparison

The scarce resource is 100 shares in an IPO-style offering. We compare two consequential allocation rules. **M1: uniform-price auction** sorts price–quantity bids and allocates until shares are exhausted; winners pay the lowest accepted price. **M2: bookbuilding** uses investor indications and underwriter discretion, with the current model assigning 85% of allocation to institutions and applying an 8% pricing discount. These are deliberately different mechanisms, not two labels for the same rule.

The key innovation is that the auction also creates variation in proceeds. Write the observed outcome as $y = \theta + \rho k + \varepsilon$. A naive learner that regresses outcome on disclosure obtains

$$\hat{b}_{naive} = b_{true} + \rho \frac{\text{Cov}(k, d)}{\text{Var}(d)} + R, \quad (5)$$

where the middle term is the evaluator’s own capital footprint and R captures remaining heterogeneity. We therefore compare three evaluator modes: **no memory**, **naive history**, and **capital-purged**

history. The last uses allocation variation as an instrument for k and is gated by a first-stage strength check.

This makes mechanism design consequential. The question is not merely which auction raises more revenue. It is whether the allocation rule supplies enough credible variation to identify the feedback that the evaluator must remove before learning from outcomes. The analogy boundary is explicit: real bookbuilding also includes certification and relationship effects that our controlled model does not represent.

5 Computational and behavioral test

The computational design uses 200 synthetic issuers, 10 cohorts, seeds 42–46, 100 shares per offering, 15 institutional and 35 retail investors, and the same disclosure, signal, memory, and auction rules described above. The primary outcomes are mean manipulation $\bar{m}$, price RMSE, capital-allocation efficiency, feedback bias $\hat{b} - b_{true}$, manipulation cost, retail allocation share, and the first-stage F statistic.

The current simulation reports, for example, $\hat{b} = 1.42$ versus $b_{true} = 1.02$ under naive memory and an increase in mean manipulation from 0.233 to 0.402. The current bookbuilding condition produces substantially stronger allocation-shock variation than the uniform-price condition. These are model outputs, not evidence about real issuers; all reported numbers must be regenerated from the final code and seed record before submission.

The behavioral artifact tests the equilibrium assumption that investors remove predictable disclosure inflation from the public AI signal. A competing explanation is anchoring or algorithm appreciation [15]. Players therefore face the same strategic setting as the model and their implied weight on the public signal is compared with the Bayesian benchmark. Classroom/Hugging Face evidence is exploratory: it can challenge or qualify the behavioral assumption but cannot establish population-level investor behavior.

Integrated contribution. Game theory predicts strategic disclosure; social choice defines the collective objectives and trade-offs; mechanism design changes the allocation rule and the information available for evaluator learning; computation tests the feedback mechanism; and behavioral play challenges the rational-investor benchmark. The practical implication is testable: any AI system that both evaluates and influences capital should expose enough of its calibration path and allocation footprint for an external auditor to distinguish genuine performance from performance it helped create.

Author Notes

Acknowledgements

Course instructor: Professor Luyao Zhang. We thank [peer reviewer names] for feedback to be recorded after the September 28 symposium and [discussant name] for questions about the identification assumptions. We build on published work by Lacker and Weinberg [8], Crocker and Slemrod [9], Cao, Jiang, Yang and Zhang [6], Benveniste and Spindt [10], Sherman [11], Degeorge, Derrien and Womack [12], Perdomo et al. [13], Staiger and Stock [14], and Logg, Minson and Moore [15]; none of these authors was involved in this work. Software used or planned includes Python 3.11, NumPy, pandas, Matplotlib, and Gradio. The ACM `acmart` class and course template are used as provided. Generative AI assistance is disclosed in Appendix A.1. Feedback or a program role does not imply coauthorship. The authors remain responsible for the proposal and remaining errors.

Team and individual contributions

Joint contribution. We formulate a single feedback problem linking strategic disclosure, AI valuation, capital allocation, and subsequent AI learning. The common contribution is the design of a setting in which an issuer knows the evaluator, the evaluator’s signal affects the auction outcome, capital changes the issuer’s later performance, and that performance becomes part of the evaluator’s history. This makes the evaluator’s own allocation footprint an object of analysis rather than treating valuation and allocation as separate modules. The team jointly specifies the three connected questions, the strategic environment, the social-planner objectives, the auction comparison, and the computational/behavioral tests. The resulting argument is: game theory predicts disclosure and bidding responses; social choice defines the trade-off among informative pricing, allocative efficiency, and access; and mechanism design changes the allocation rule and therefore the information available for safe history-based calibration.

Feiyu Li (author 1). *Personal intellectual decision:* reframing the project around the joint design problem of an AI evaluator and a capital-allocation mechanism, with particular attention to the distinction between genuine firm quality and performance induced by capital allocated through the evaluator. *Inspectable output:* the strategic-disclosure formulation, issuer-type specification, auction interface, behavioral-game design, and Sections 1, 2, 4, and 5. *Verification and result:* independently re-ran the stated mechanism-by-memory comparison on seeds 42–46 and checked the sign and magnitude of the reported change in manipulation under naive historical learning. Feiyu also checked the investor-side interpretation of the public signal against the Bayesian benchmark. *Connection:* this work connects the firm’s disclosure incentive to the auction and to the behavioral test rather than treating them as separate examples. *Next responsibility:* after symposium feedback, verify the final computational implementation against the behavioral interface and document any revision to the evaluator or auction assumptions.

Xuantong Fu (author 2). *Personal intellectual decision:* identifying the evaluator’s own capital allocation as the source of a feedback/omitted-variable problem and making the allocation mechanism consequential for whether that footprint can be separated

from true firm quality. *Inspectable output:* the memory/update estimators, allocation-shock identification argument, auction comparison, bias decomposition, and supporting technical derivations in Section 3 and Appendix A. *Verification and result:* checked the analytic footprint component against the simulated allocation-induced outcome and verified the first-stage diagnostic used to decide whether capital-purged estimation is informative. *Connection:* this identification work turns the auction comparison into part of the evaluator-design problem: a mechanism is useful not only because it allocates shares, but also because its variation determines what the evaluator can safely learn from. *Next responsibility:* complete the robustness checks for the allocation-shock assumptions and report whether the proposed purging procedure remains valid under alternative signal and allocation parameters.

Documented cross-check. Each author independently verifies a consequential component of the other’s work. Feiyu checks the computational outputs and behavioral-interface mapping from a clean checkout; Xuantong recomputes the core manipulation, price-informativeness, and allocation-efficiency measures and checks that their interpretation matches the stated social objective. Both authors review the consistency of the theory, simulation, behavioral artifact, and paper before submission.

6 Technical details and reproducibility

Separating-equilibrium benchmark. With $c(m) = m^2/2\gamma$ and a linear price response, the cost of increasing the report is convex. The single-crossing condition supports a separating benchmark in which the optimal manipulation is increasing in the evaluator’s perceived sensitivity to disclosure. The exact expression used in the simulation is

$$m^* = \gamma \nu n \lambda_P [\lambda \beta b + (1 - \lambda) \kappa_0]. \quad (6)$$

This is the strategic link between evaluator design and disclosure cost; it is not interpreted as an empirical estimate of any real issuer.

Feedback decomposition. The simulated learning problem can be written as

$$\hat{b} - b_{\text{true}} = \rho \frac{\text{Cov}(k, d)}{\text{Var}(d)} + R, \quad (7)$$

where the first term is the capital footprint targeted by the allocation-shock correction and R collects remaining heterogeneity. In the current model the footprint is generated by the mechanism; therefore the correction is only as credible as the instrument assumptions. The unresolved residual is explicitly treated as a limitation and planned robustness target rather than as a solved empirical problem.

Language-independent pseudocode.

```
for seed in {42,...,46}:
  initialize issuer types and private signals
  for issuer in sequential_IPOs:
    choose disclosure d from strategic best response
    compute public AI signal s from memory mode
    collect investor bids from private + public information
    P, allocation, psi <- run_auction(bids, mechanism)
    k <- proceeds_per_share(P, psi)
    y <- theta + rho*k + noise
    update evaluator memory using mode
  aggregate RMSE, manipulation, allocation efficiency,
  price error, welfare, retail access, and bias diagnostics
```

Inputs and parameters. The computational design uses $N = 200$ issuers (10 cohorts $\times$ 20), 100 shares per offering, seeds 42–46, a two-cohort burn-in, $\sigma_\theta = 1.0$, $\gamma = 0.006$, $\sigma_\eta = 0.5$, $\mu_0 = 8.5$, $\beta = 0.60$, $\lambda = 1.0$, and $\sigma_\epsilon = 0.8$. The investor pool contains 15 institutions ($\sigma_\xi = 1.0$) and 35 retail investors ($\sigma_\xi = 2.5$). The allocation-shock scale differs by mechanism. Full values should be maintained in the computational configuration file and echoed into the fresh-run record.

Metrics. Primary outcomes are (i) manipulation, $\bar{m}$ and manipulation cost; (ii) price informativeness, measured by RMSE between the transaction price and the latent fundamental value; (iii) capital-allocation efficiency, measured by the relationship between capital and ρ ; and (iv) evaluator feedback bias, $\hat{b} - b_{\text{true}}$, with its footprint decomposition. Secondary outcomes include retail allocation share, realised investor return, and the first-stage F statistic for the allocation-shock correction.

Known limitations. (1) ρ is exogenous; the model does not solve a dynamic general-equilibrium problem. (2) Memory is cross-firm rather than cross-period, so the model is an abstraction of repeated calibration rather than a literal representation of seasoned

issuance. (3) The AI is represented as a rule rather than as a strategic vendor with its own payoff. (4) Only one evaluator is modelled; competing valuation systems are left for future work. (5) The heterogeneity residual R in Eq. (7) is unresolved. (6) All data are synthetic; no claim is made about any real issuer. (7) Behavioral evidence is exploratory and cannot establish population-level investor behavior.

Actual versus planned evidence. The numerical values currently labelled “actual” are outputs of the team’s stated simulation design. They should remain labelled actual only after the code, seed list, and fresh-run record are checked from a clean checkout. The classroom/Hugging Face behavioral test, final poster, and post-symposium review response are planned or pending until those activities occur. No future activity is represented as completed here.

Fresh-run record. GitHub commit [COMMIT_SHA], release [TAG]; notebook notebooks/PS2_demo.ipynb; Space revision [HF_SHA]; poster version [POSTER_VERSION]. The final record should include the UTC timestamp, Python version, package versions, seed list, and the exact command used to regenerate the reported outputs.

6.1 AI-use disclosure

[Complete before submission. Record tool/model, date, purpose, significant prompts, suggestions accepted or rejected, human changes, and verification. The current project used generative AI for drafting and debugging only after the team had independently specified the research question, hypotheses, and core model. Any generated text or code must be checked by the human authors. If no generative AI was used for a component, state that explicitly.]

7 Cumulative Intellectual Development

Prior methodological starting point. The project began from the course’s strategic-disclosure setting. The PS2 revision adds a consequential allocation mechanism and a history-learning loop rather than treating the evaluator as a static scoring rule.

Revision path. The key change was to make the evaluator’s output affect capital allocation and then make realised performance feed back into the evaluator. This created a testable distinction between genuine information in disclosure and performance induced by the evaluator’s own decisions. The auction comparison was then made part of the identification problem rather than a decorative application.

Intellectual consequence. Asking whether the evaluator learns from outcomes it caused changed the central question from “can firms manipulate the AI?” to “how should an evaluator learn when its own pricing changes the data-generating process?” This change affects the strategic model, social objective, mechanism comparison, computational metrics, and behavioral interface.

8 Open-source and three-lens inspiration

Strategic/game-theoretic foundation. Harsanyi [2] supplies the incomplete-information equilibrium logic, while costly-state-falsification models [8, 9] motivate strategic disclosure. These foundations lead us to model private issuer types, disclosure costs, sequential moves, and beliefs explicitly.

Social-choice foundation. The social planner does not have a single objective. Informative prices, efficient allocation of scarce capital, disclosure costs, and access for retail investors can conflict. The project therefore treats welfare as a multi-objective design problem and reports the trade-off rather than collapsing it into one accuracy statistic.

Mechanism-design foundation. Myerson [4] and the auction literature motivate comparing allocation/payment rules. The consequential design decision here is whether the mechanism creates usable variation in proceeds without changing the issuer’s disclosure decision directly. This is why the auction comparison enters the evaluator-learning problem.

Computational and behavioral integration. GitHub makes the simulation reconstructable; Hugging Face makes the behavioral assumption about investor reliance observable; the A0 poster must integrate the same question, evidence, limitations, and real-world case. Behavioral play is treated as a challenge test of the equilibrium benchmark, not as population validation.

9 Review response and revision record

Review-ready status. This working version is prepared for the September 28 symposium. No symposium review is represented as completed before the event.

Source	Feedback, decision, location, rerun, result
Peer 1	[To complete after review: evidenced strength; priority concern; Keep/Revise/Investigate; changed section/file; verification; result.]
Peer 2	[To complete after review.]
Discussant	[To complete after review.]
Instructor	[To complete after review.]

Pre-specified challenge. The exogeneity of the allocation shock ψ is load-bearing. In the model it is imposed by construction; in an empirical application it would need independent justification because rationing or underwriter discretion could correlate with unobserved demand and quality. We therefore classify empirical validation of the instrument as **Investigate**, not as a demonstrated real-world result. The final response will record the post-symposium decision and the corresponding verification.

10 Structured abstract and cumulative proposal map

Table 1: Structured abstract, PS2.

Proposal element	Current statement / change
Background	Investors increasingly use automated valuation signals built from issuer disclosures. Strategic disclosure can distort what machine readers observe [6]; costly-state-falsification models study such incentives under fixed rules [8, 9]. The unresolved setting here is a rule that both evaluates and affects capital allocation, so its later training data can partly reflect its own decisions.
Motivation and application	IPO-style capital allocation is used as a controlled mechanism-design environment. Stakeholders include issuers, institutional investors, retail investors, exchanges, regulators, and later issuers who inherit the evaluator’s calibration.
Three connected questions	<i>Economics</i> : how does strategic disclosure respond to a known evaluator? <i>Computation</i> : can evaluator learning separate true quality from allocation-induced performance? <i>Behavior</i> : do investors remove predictable disclosure inflation from the public signal? <i>Integrated</i> : how should evaluator design and allocation design be jointly chosen?
Method and model assumptions	Private issuer types (θ, v, ρ) ; sequential disclosure, valuation, bidding, allocation, and outcome; PBE [2]; no-memory, naive-history, and capital-purged memory modes; uniform-price auction versus bookbuilding; 200 synthetic issuers and seeds 42–46.
Results or expected results	Current simulation output : naive memory reports $\hat{b} = 1.42$ versus $b_{\text{true}} = 1.02$ and increases mean manipulation from 0.233 to 0.402 in the current run. Bookbuilding provides stronger allocation-shock variation in the reported simulation than the uniform-price condition. Planned verification : regenerate all values from a clean checkout and test robustness to alternative signal noise, allocation shocks, and heterogeneous capital productivity.
Intellectual merits	The contribution is the feedback loop: the evaluator’s signal affects capital, capital affects outcomes, and outcomes enter future calibration. The auction is consequential because its allocation variation determines whether that feedback can be separated from genuine disclosure information.
Practical impacts	A possible design standard is to require an AI valuation rule that also allocates capital to disclose its calibration path and the evidence separating evaluator-induced outcomes from genuine quality. Benefits, risks, access constraints, and correction routes remain design hypotheses until validated on external data.
Feedback received and revision made	The PS1/Week 4 starting point treated buyer behavior too mechanically. The PS2 revision makes investor bidding strategic and makes the auction consequential for both allocation and evaluator learning. The September 28 symposium is the next formal challenge point; its feedback will be recorded after the event rather than anticipated as completed.

11 PS2 extension record: auction, linked artifacts, poster, and symposium

11.1 Auction laboratory record

Rules, stated explicitly. *Uniform-price (M1).* Allocation: bids are sorted descending and accepted until the 100-share offering is exhausted. Payment: winners pay the lowest accepted price. Stopping: cumulative demand reaches the offering. *Bookbuilding (M2).* Allocation: underwriter discretion with the current model assigning 85% to institutions. Payment: the model uses an underwriter price with an 8% discount to the precision-weighted institutional estimate. Stopping: close of the book. The analogy boundary is explicit: real IPO bookbuilding also involves certification, relationship, and allocation considerations not represented here.

Treatment	Setting, prediction, current output, revision status
M1, no memory	Baseline. Predicted: anticipated disclosure inflation can be stripped from the public signal. Current output: RMSE 1.85, underpricing 7.1%, retail share 54.6%, waste 0.6%.
M1, naive history	Predicted: learning from the outcome history may amplify the perceived value of disclosure. Current output: $\hat{b} = 1.42$, mean manipulation 0.402, waste 1.7%.
M1, purged history	Predicted: removing the capital footprint reduces feedback bias when the allocation shock is strong enough. Current output: first-stage $F = 17$; this is a weak-instrument boundary case and is treated as a limitation rather than a success claim.
M2, no memory	Baseline bookbuilding condition. Current output: RMSE 1.82, retail share 15.0%, waste 0.6%.
M2, naive history	Current output: $\hat{b} = 1.42$, mean manipulation 0.398, waste 1.6%.
M2, purged history	Predicted: stronger allocation discretion supplies more useful variation. Current output: first-stage $F = 2354$ and lower reported feedback bias. Robustness verification remains required.
Classroom/HF	Planned September 28 behavioral test: record game URL/version, settings, own signal/value, planned action, equilibrium prediction, actual bid/payment/utility/allocation, behavioral observation, and resulting revision.

Behavioral observation to record. The main test is whether players remove predictable disclosure inflation from the AI signal. Algorithm appreciation [15] and simple anchoring make competing predictions about how the weight on the public signal changes when the evaluator’s calibration status is disclosed. The result will be labelled exploratory.

11.2 Linked-artifact parity

Artifact	Parity check
GitHub	Must implement the same issuer types, disclosure best response, memory modes, auction rules, parameters, and seeds as the paper. A clean run must regenerate every reported numerical table.
Hugging Face	Must expose the same decision setting used in the behavioral section: strategic disclosure, AI valuation, investor action, and allocation. The interface must not imply population-level behavioral validity.
A0 poster	Must use the same research question, core model, auction comparison, verified numerical outputs, limitations, real-world case, and correction route as the paper.

11.3 Symposium and practical-impact record

Presentation version. Planned for September 28, 2026, IB 2050. Poster version and artifact revisions will be recorded after the working links are frozen.

Real-world case record. The application is AI-assisted IPO/equity valuation and allocation. Affected stakeholders include genuine capital-constrained issuers, weaker issuers, institutional investors, retail investors, exchanges, regulators, and later issuers affected by the evaluator’s history. Potential benefit: an auditable criterion for deciding when a valuation rule can safely learn from outcomes. Risk: a naive self-calibrated rule may look strong on backtests precisely because its errors track its own past allocations. Missing evidence: real issuance data with defensible allocation shocks. Correction route: publish the evaluator’s calibration path, memory-update log, and identification diagnostics so an outside auditor can challenge the feedback assumption.

Final revision record. To be completed after the symposium: for each consequential review comment, record Keep/Revise/Investigate, the section or artifact changed, the verification performed, and the resulting output.

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